

Tharun C

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Profile

Final-year B.Tech (AI & Machine Learning) undergrad building production-grade ML systems and full-stack applications, from fine-tuning transformer models to designing REST APIs and deploying ML microservices end-to-end. Strong communication and teamwork skills, comfortable working in Agile, cross-functional teams from prototype to production.

Skills

Languages: Java, Python, JavaScript, TypeScript, SQL

Web/Backend: HTML, CSS, REST APIs, Node.js, Express, FastAPI, Next.js

Databases: PostgreSQL (SQL), Firebase/Firestore (NoSQL)

AI/ML: PyTorch, HuggingFace Transformers, Scikit-Learn, CodeBERT Fine-Tuning, ONNX, Prompt Engineering

Frontend: React, React Native, TypeScript, Tailwind CSS

Tools: Git, GitHub Actions (CI/CD), AWS, Agile/Scrum

Projects

Neural Vulnerability Detector – *Fine-Tuned CodeBERT for Security Analysis* — Python, PyTorch, FastAPI, ONNX, CUDA [Link](#)

- Achieved F1 0.782, recall 0.722, and precision 0.853 by fine-tuning a 124M-parameter CodeBERT transformer on 190,000+ CVE functions (BigVul) – catching 741/1,026 vulnerabilities with only 128 false alarms across 32,024 clean functions.
- Fixed a 16:1 class imbalance with weighted cross-entropy loss and a tuned decision threshold (0.9985), boosting recall without inflating false positives.
- Cut inference to 250ms via ONNX export, then shipped it 3 ways: a FastAPI server, a GitHub Action auto-commenting CWE issues on PRs, and a VS Code extension.

Tripzo – *AI-Powered Ride Booking Platform* — React Native, Node.js, FastAPI, Scikit-Learn, PostgreSQL [Link](#)

- Built a full-stack ride-hailing app (Expo + TypeScript + Zustand) with a Scikit-Learn fare-prediction microservice trained on 7 real-world factors (distance, traffic, weather, demand, time of day).
- Replaced static, straight-line pricing with live road-distance and traffic/weather-aware fares via 4 Google Maps APIs, plus a transparent breakdown engine with surge up to 2.0x and a confidence score.
- Secured payments with Razorpay signature verification and Clerk auth across a Node.js/Express REST API on a normalized PostgreSQL schema (NeonDB), following Agile sprints for iterative feature delivery.

InterviewX – *AI Voice-Driven Mock Interview Platform* — Next.js, Gemini API, Deepgram, Firebase [Link](#)

- Built a voice-driven mock interview platform (Next.js 15) generating on-demand questions via Gemini 2.5 Flash, scaling difficulty across 3 levels through prompt design alone.
- Enforced per-user data isolation at the query level with a 3-collection Firestore (NoSQL) schema, secured by HTTP-only session cookies via the Firebase Admin SDK.
- Built a resilient voice pipeline (Deepgram STT → Gemini → TTS) that sanitizes malformed LLM JSON and falls back gracefully on AI failure.

Certifications

- **Foundations of Deep Learning: Concepts and Applications** – NPTEL (IIT/IISc, Govt. of India), Elite, 82% (2026)
- **AWS Academy Graduate – Cloud Foundations** (2025)
- **IBM SkillsBuild – Cloud Computing Fundamentals** (2025)

Education

B.Tech in Artificial Intelligence & Machine Learning

Dayananda Sagar University

Expected 2027
Bengaluru, India

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Deep Learning